

Chapter 1

The Foundations: Logic and Proofs

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Chapter Summary

Propositional Logic (1.1-1.3)

- The Language of Propositions
- Applications
- Logical Equivalences

Predicate Logic (1.4)

- The Language of Quantifiers
- Logical Equivalences

Proofs (1.6-1.7)

- Rules of Inference
- Proof Methods

Propositional Logic

Section 1.1

Propositional Logic Summary

The Language of Propositions

- Connectives
- Truth Values
- Truth Tables

Applications

- Translating English Sentences
- System Specifications
- Logic Puzzles
- Logic Circuits

Logical Equivalences

- Important Equivalences
- Showing Equivalence
- Satisfiability

Section Summary

Propositions

Connectives

- Negation
- Conjunction
- Disjunction
- Implication; contrapositive, inverse, converse
- Biconditional

Truth Tables

Propositions

A *proposition* is a declarative sentence that is **either true or false**.

Are these propositions?

a) The Moon is made of green cheese.

b) Trenton is the capital of New Jersey.

c) Toronto is the capital of Canada.

d) $1 + 0 = 1$

e) $0 + 0 = 2$

f) Sit down! 

g) What time is it? 

h) $x + 1 = 2$ 

i) $x + y = z$ 

Propositional Logic

Constructing Propositions

- Propositional Variables: $p, q, r, s, \dots$
- The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.
- Compound Propositions - constructed from logical connectives and other propositions
 - Negation $\neg$
 - Conjunction $\wedge$
 - Disjunction $\vee$
 - Implication $\rightarrow$
 - Biconditional $\leftrightarrow$

Compound Propositions: Negation

The *negation* of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

Example: If p denotes “The earth is round.”, then $\neg p$ denotes “It is not the case that the earth is round,” or more simply “The earth is not round.”

Conjunction

The *conjunction* of propositions p and q is denoted by $p \wedge q$ and has this truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes

“I am at home and it is raining.”

Disjunction

The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes

“I am at home or it is raining.”

The Connective Or in English

In English “or” has two distinct meanings.

- “Inclusive Or” - In the sentence “Students who have taken CS202 or Math120 may take this class,” we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of disjunction. For $p \vee q$ to be true, either one or both of p and q must be true.
- “Exclusive Or” - When reading the sentence “Soup or salad comes with this entrée,” we do not expect to be able to get both soup and salad. This is the meaning of Exclusive Or (Xor). In $p \oplus q$, one of p and q must be true, but not both. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication

If p and q are propositions, then $p \rightarrow q$ is a **conditional statement** or **implication** which is read as “if p , then q ” and has this truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Example: If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “If I am at home then it is raining.”

In $p \rightarrow q$, p is the *hypothesis* (*antecedent* or *premise*) and q is the *conclusion* (or *consequence*).

Understanding Implication₁

In $p \rightarrow q$ there does not need to be any connection between the antecedent or the consequent. The “meaning” of $p \rightarrow q$ depends only on the truth values of p and q .

These implications are perfectly fine, but would not be used in ordinary English.

- “If the moon is made of green cheese, then I have more money than Bill Gates. ”
- “If the moon is made of green cheese then I’m on welfare.”
- “If $1 + 1 = 3$, then your grandma wears combat boots.”

Understanding Implication₂

One way to view the logical conditional is to think of an obligation or contract.

- “If I am elected, then I will lower taxes.”
- “If you get 100% on the final, then you will get an A.”

If the politician is elected and does not lower taxes, then the voters can say that he or she has broken the campaign pledge. Something similar holds for the professor. This corresponds to the case where p is true and q is false.

Different Ways of Expressing $p \rightarrow q$

if p , then q

p implies q

if p , q

p only if q

q unless $\neg p$

q when p

q if p

q whenever p

p is sufficient for q

q follows from p

q is necessary for p

a necessary condition for p is q

a sufficient condition for q is p

Converse, Contrapositive, and Inverse

From $p \rightarrow q$ we can form new conditional statements .

- $q \rightarrow p$ is the **converse** of $p \rightarrow q$
- $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$
- $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It is raining is a sufficient condition for me not going to town.”

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

Biconditional

If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as “ p if and only if q .” The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”

Expressing the Biconditional

Some alternative ways “ p if and only if q ” is expressed in English:

- p is necessary and sufficient for q
- if p then q , and conversely
- p if q

Truth Tables For Compound Propositions

Construction of a truth table:

Rows

- Need a row for every possible combination of values for the atomic propositions.

Columns

- Need a column for the compound proposition (usually at far right)
- Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
 - This includes the atomic propositions

Example Truth Table

Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

Two propositions are *equivalent* if they always have the same truth value.

Example: Show using a truth table that the conditional is equivalent to the contrapositive.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Problem

How many rows are there in a truth table with n propositional variables?

Solution: 2^n

Note that this means that with n propositional variables, we can construct 2^n distinct (that is, not equivalent) propositions.

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$

If the intended meaning is $p \vee (q \rightarrow \neg r)$

then parentheses must be used.

Applications of Propositional Logic

Section 1.2

Section Summary

Translating English to Propositional Logic

System Specifications

Logic Puzzles

Logic Circuits

Translating English Sentences

Steps to convert an English sentence to a statement in propositional logic

- Identify atomic propositions and represent using propositional variables.
- Determine appropriate logical connectives

“If I go to school or to the gym, I will not go shopping.”

- p : I go to school
- q : I go to the gym
- r : I will go shopping

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

One Solution: Let a , c , and f represent respectively “You can access the internet from campus,” “You are a computer science major,” and “You are a freshman.”

$$a \rightarrow (c \vee \neg f)$$

System Specifications

System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply cannot be sent when the file system is full”

Solution: One possible solution: Let p denote “The automated reply can be sent” and q denote “The file system is full.”

$$q \rightarrow \neg p$$

Consistent System Specifications

Definition: A list of propositions is **consistent** if it is possible to assign truth values to the proposition variables so that each proposition is true.

Exercise: Are these specifications consistent?

- “The diagnostic message is stored in the buffer or it is retransmitted.”
- “The diagnostic message is not stored in the buffer.”
- “If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution: Let p denote “The diagnostic message is stored in the buffer.” Let q denote “The diagnostic message is retransmitted” The specification can be written as: $p \vee q, \neg p, p \rightarrow q$. When p is false and q is true all three statements are true. So the specification is consistent.

- What if “The diagnostic message is not retransmitted” is added.

Solution: Now we are adding $\neg q$ and there is no satisfying assignment. So the specification is not consistent.

Logic Puzzles

An island has two kinds of inhabitants, *knight*s, who always tell the truth, and *knave*s, who always lie.

You go to the island and meet A and B.

- A says “B is a knight.”
- B says “The two of us are of opposite types.”

Example: What are the types of A and B?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

- 1) **If A is a knight**, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- 2) **If A is a knave**, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

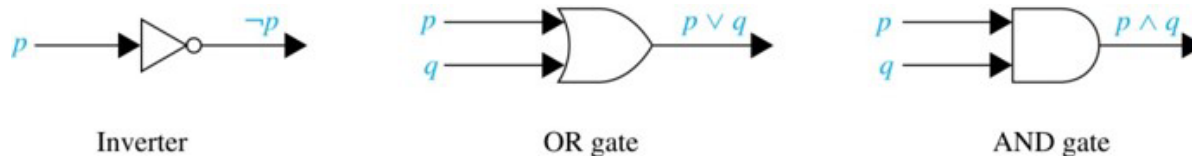
Logic Circuits

(Studied in depth in Chapter 12)

Electronic circuits; each input/output signal can be viewed as a 0 or 1.

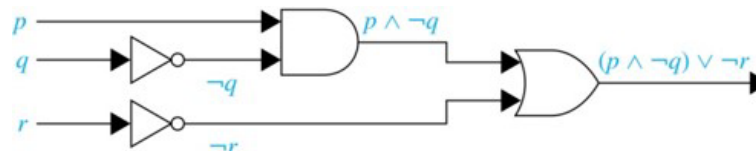
- 0 represents **False**
- 1 represents **True**

Complicated circuits are constructed from three basic circuits called gates.



- The inverter (**NOT gate**) takes an input bit and produces the negation of that bit.
- The **OR gate** takes two input bits and produces the value equivalent to the disjunction of the two bits.
- The **AND gate** takes two input bits and produces the value equivalent to the conjunction of the two bits.

More complicated digital circuits can be constructed by combining these basic circuits to produce the desired output given the input signals by building a circuit for each piece of the output expression and then combining them. For example:



Propositional Equivalences

Section 1.3

Section Summary

Tautologies, Contradictions, and Contingencies.

Logical Equivalence

- Important Logical Equivalences
- Showing Logical Equivalence (Proof)

Propositional Satisfiability

Tautologies, Contradictions, and Contingencies

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

A **tautology** is a proposition which is always true.

- Example: $p \vee \neg p$

A **contradiction** is a proposition which is always false.

- Example: $p \wedge \neg p$

A **contingency** is a proposition which is neither a tautology nor a contradiction, such as p

Logically Equivalent

Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.

We write this as $p \leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.

Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.

This truth table shows that $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$



Augustus De Morgan
1806-1871

Use this truth table to show that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences₁

Identity Laws: $p \wedge T \equiv p, \quad p \vee F \equiv p$

Domination Laws: $p \vee T \equiv T, \quad p \wedge F \equiv F$

Idempotent laws: $p \vee p \equiv p, \quad p \wedge p \equiv p$

Double Negation Law: $\neg(\neg p) \equiv p$

Negation Laws: $p \vee \neg p \equiv T, \quad p \wedge \neg p \equiv F$

Key Logical Equivalences₂

Commutative Laws: $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$

Associative Laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
 $(p \vee q) \vee r \equiv p \vee (q \vee r)$

Distributive Laws: $(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$
 $(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$

Absorption Laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

More Logical Equivalences*

TABLE 7 Logical Equivalences
Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences
Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

* Provided for exams

Constructing New Logical Equivalences

We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.

To *prove* that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B .

$$\begin{array}{c} A \equiv A_1 \\ \vdots \\ A_n \equiv B \end{array}$$

Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs₁

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
$\equiv \neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
$\equiv \neg p \wedge (p \vee \neg q)$	by the double negation law
$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
$\equiv F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
$\equiv (\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
$\equiv (\neg p \wedge \neg q)$	By the identity law for F

Equivalence Proofs₂

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology.

Solution:

$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$	by logical equivalency for $\rightarrow$
$\equiv (\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
$\equiv (\neg p \vee p) \vee (\neg q \vee q)$	by associative and commutative laws for disjunction
$\equiv T \vee T$	by truth tables
$\equiv T$	by the domination law

Propositional Satisfiability

A compound proposition is *satisfiable* if there is an assignment of truth values to its variables that make it true. When no such assignments exist, the compound proposition is *unsatisfiable*.

A compound proposition is unsatisfiable if and only if its negation is a tautology.

Propositional Satisfiability

Example: Determine the satisfiability of the following compound propositions:

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$$

Solution: Satisfiable. Assign **T** to p , q , and r .

$$(p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Satisfiable. Assign **T** to p and **F** to q .

$$(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p) \wedge (p \vee q \vee r) \wedge (\neg p \vee \neg q \vee \neg r)$$

Solution: Not satisfiable. Check each possible assignment of truth values to the propositional variables and none will make the proposition true.

Predicates and Quantifiers

Section 1.4

Section Summary

Predicates

Variables

Quantifiers

- Universal Quantifier
- Existential Quantifier

Negating Quantifiers

- De Morgan's Laws for Quantifiers

Translating English to Logic

Propositional Logic Not Enough

If we have:

“All men are mortal.”

“Socrates is a man.”

Does it follow that “Socrates is mortal?”

Can't be represented in propositional logic.

Need a language that talks about objects, their properties, and their relations.

Later we'll see how to draw inferences.

Introducing Predicate Logic

Predicate logic uses the following new features:

- Variables: x, y, z
- Predicates: $P(x), M(x)$
- Quantifiers (*to be covered in a few slides*):

Propositional functions are a generalization of propositions.

- They contain variables and a predicate, e.g., $P(x)$
- Variables can be replaced by elements from their *domain*.

Propositional Functions

Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the *domain* (or *bound* by a quantifier, as we will see later).

The statement $P(x)$ is said to be the value of the propositional function P at x .

For example, let $P(x)$ denote “ $x > 0$ ” and the domain be the integers. Then:

$P(-3)$ **F**

$P(0)$ **F**

$P(3)$ **T**

Often the domain is denoted by U . So in this example U is the integers.

Examples of Propositional Functions

Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ and U (for all three variables) be the integers. Find these truth values:

$R(2, -1, 5)$

Solution: **F**

$R(3, 4, 7)$

Solution: **T**

$R(x, 3, z)$

Solution: **Not a Proposition**

Now let “ $x - y = z$ ” be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:

$Q(2, -1, 3)$

Solution: **T**

$Q(3, 4, 7)$

Solution: **F**

$Q(x, 3, z)$

Solution: **Not a Proposition**

Compound Expressions

Connectives from propositional logic carry over to predicate logic.

If $P(x)$ denotes “ $x > 0$,” find these truth values:

$P(3) \vee P(-1)$ **Solution:** **T**

$P(3) \wedge P(-1)$ **Solution:** **F**

$P(3) \rightarrow P(-1)$ **Solution:** **F**

$P(3) \rightarrow \neg P(-1)$ **Solution:** **T**

Expressions with variables are not propositions and therefore **do not** have truth values. For example,

$P(3) \wedge P(y)$

$P(x) \rightarrow P(y)$

When used with quantifiers (to be introduced next), these expressions (propositional functions) become propositions.

Quantifiers

We need *quantifiers* to express the meaning of English words including ***all*** and ***some***:

- “All men are Mortal.”
- “Some cats do not have fur.”

The two most important quantifiers are:

- *Universal Quantifier*, “For all,” symbol: $\forall$
- *Existential Quantifier*, “There exists,” symbol: $\exists$

We write as in $\forall x P(x)$ and $\exists x P(x)$.

$\forall x P(x)$ asserts $P(x)$ is true for every x in the *domain*.

$\exists x P(x)$ asserts $P(x)$ is true for some x in the *domain*.

The quantifiers are said to bind the variable x in these expressions.

Universal Quantifier

$\forall x P(x)$ is read as “For all x , $P(x)$ ” or “For every x , $P(x)$ ”

Examples:

- 1) If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\forall x P(x)$ is **False**.
- 2) If $P(x)$ denotes “ $x > 0$ ” and U is the positive integers, then $\forall x P(x)$ is **True**.
- 3) If $P(x)$ denotes “ x is even” and U is the integers, then $\forall x P(x)$ is **False**.

Existential Quantifier

$\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples:

1. If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
2. If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is **False**.
3. If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is **True**.

Thinking about Quantifiers

When the domain of discourse is finite, we can think of quantification as looping through the elements of the domain.

To evaluate $\forall x P(x)$, loop through all x in the domain.

- If at every step $P(x)$ is true, then $\forall x P(x)$ is true.
- If at a step $P(x)$ is false, then $\forall x P(x)$ is false and the loop terminates.

To evaluate $\exists x P(x)$, loop through all x in the domain.

- If at some step, $P(x)$ is true, then $\exists x P(x)$ is true and the loop terminates.
- If the loop ends without finding an x for which $P(x)$ is true, then $\exists x P(x)$ is false.

Even if the domains are infinite, we can still think of the quantifiers this fashion, but the loops will not terminate in some cases.

Properties of Quantifiers

The truth value of $\exists x P(x)$ and $\forall x P(x)$ depend on both the propositional function $P(x)$ and on the domain U .

Examples:

1. If U is the positive integers and $P(x)$ is the statement “ $x < 2$ ”, then $\exists x P(x)$ is **T**, $\forall x P(x)$ is **F**.
2. If U is the negative integers and $P(x)$ is the statement “ $x < 2$ ”, then $\exists x P(x)$ is **T**, $\forall x P(x)$ is **T**.
3. If U consists of 3, 4, and 5, and $P(x)$ is the statement “ $x > 2$ ”, then both $\exists x P(x)$ and $\forall x P(x)$ are true. But if $P(x)$ is the statement “ $x < 2$ ”, then both $\exists x P(x)$ and $\forall x P(x)$ are false.

Precedence of Quantifiers

The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.

For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$

$\forall x (P(x) \vee Q(x))$ means something different.

Unfortunately, often people write $\forall x P(x) \vee Q(x)$ when they mean $\forall x (P(x) \vee Q(x))$.

Translating from English to Logic

Example 1: Translate the following sentence into predicate logic: “Every student in this class plays League of Legends (LOL).”

Solution:

First decide on the domain U .

Solution 1: If U is all students in this class, define a propositional function $L(x)$ denoting “ x plays LOL” and translate as $\forall x L(x)$.

Solution 2: But if U is all people, also define a propositional function $S(x)$ denoting “ x is a student in this class” and translate as $\forall x (S(x) \rightarrow L(x))$.

$\forall x (S(x) \wedge L(x))$ is not correct. What does it mean?

Translating from English to Logic₂

Example 2: Translate the following sentence into predicate logic: “Some student in this class play LOL.”

Solution:

First decide on the domain U .

Solution 1: If U is all students in this class, translate as

$$\exists x L(x)$$

Solution 2: But if U is all people, then translate as

$$\exists x (S(x) \wedge L(x))$$

$\exists x (S(x) \rightarrow L(x))$ is not correct. What does it mean?

Equivalences in Predicate Logic

Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value

- for every predicate substituted into these statements and
- for every domain of discourse used for the variables in the expressions.

The notation $S \equiv T$ indicates that S and T are logically equivalent.

Example: $\forall x \neg\neg S(x) \equiv \forall x S(x)$

Thinking about Quantifiers as Conjunctions and Disjunctions

If the domain is finite, a universally quantified proposition is equivalent to a conjunction of propositions without quantifiers and an existentially quantified proposition is equivalent to a disjunction of propositions without quantifiers.

If U consists of the integers 1,2, and 3:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

Negating Quantified Expressions₁

Consider $\forall x J(x)$

“Every student in your class has taken a course in Java.”

Here $J(x)$ is “ x has taken a course in Java” and
the domain is students in your class.

Negating the original statement gives “It is not the case that every student in your class has taken Java.” This implies that “There is a student in your class who has not taken Java.”

Symbolically $\neg \forall x J(x)$ and $\exists x \neg J(x)$ are equivalent

Negating Quantified Expressions₂

Now Consider $\exists x J(x)$

“There is a student in this class who has taken a course in Java.”

Where $J(x)$ is “x has taken a course in Java.”

Negating the original statement gives “It is not the case that there is a student in this class who has taken Java.” This implies that “Every student in this class has not taken Java”

Symbolically $\neg \exists x J(x)$ and $\forall x \neg J(x)$ are equivalent

De Morgan's Laws for Quantifiers

The rules for negating quantifiers are:

TABLE 2 De Morgan's Laws for Quantifiers.			
<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

The reasoning in the table shows that:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

These are important. You will use these.

Translation from English to Logic

Examples:

1. “Some student in this class has visited Mexico.”

Solution: Let $M(x)$ denote “ x has visited Mexico” and $S(x)$ denote “ x is a student in this class,” and U be all people.

$$\exists X (S(X) \wedge M(X))$$

2. “Every student in this class has visited Canada or Mexico.”

Solution: Add $C(x)$ denoting “ x has visited Canada.”

$$\forall X (S(X) \rightarrow (M(X) \vee C(X)))$$

Translation from English to Logic

Examples: Find the negations of the followings:

1. “There is an honest politician.”

Let $H(x)$ denote “ x is honest”

$$\exists x H(x) \quad \neg \exists x H(x) \equiv \forall x \neg H(x)$$

2. “All Americans eat cheeseburgers.”

Let $C(x)$ denote “ x eats cheeseburgers”

$$\forall x C(x) \quad \neg \forall x C(x) \equiv \exists x \neg C(x)$$

System Specification Example

Predicate logic is used for specifying properties that systems must satisfy.

For example, translate into predicate logic:

- “Every mail message larger than one megabyte will be compressed.”
- “If a user is active, at least one network link will be available.”

Decide on predicates and domains (left implicit here) for the variables:

- Let $L(m, y)$ be “Mail message m is larger than y megabytes.”
- Let $C(m)$ denote “Mail message m will be compressed.”
- Let $A(u)$ represent “User u is active.”
- Let $S(n, x)$ represent “Network link n is state x .”

Now we have:

$$\forall m (L(m, 1) \rightarrow C(m))$$
$$\exists u A(u) \rightarrow \exists n S(n, \text{available})$$

Nested Quantifiers

Section 1.4

Nested Quantifiers

Nested quantifiers are often necessary to express the meaning of sentences in English as well as important concepts in computer science and mathematics.

Example: “Every real number has an inverse” is

$$\forall x \exists y (x + y = 0)$$

where the domains of x and y are the real numbers.

We can also think of nested propositional functions:

$\forall x \exists y (x + y = 0)$ can be viewed as $\forall x Q(x)$ where $Q(x)$ is $\exists y P(x, y)$ where $P(x, y)$ is $(x + y = 0)$

Thinking of Nested Quantification

Nested Loops

- To see if $\forall x \forall y P(x,y)$ is true, loop through the values of x :
 - At each step, loop through the values for y .
 - If for some pair of x and y , $P(x,y)$ is false, then $\forall x \forall y P(x,y)$ is false and both the outer and inner loop terminate.

$\forall x \forall y P(x,y)$ is true if the outer loop ends after stepping through each x .

- To see if $\forall x \exists y P(x,y)$ is true, loop through the values of x :
 - At each step, loop through the values for y .
 - The inner loop ends when a pair x and y is found such that $P(x,y)$ is true.
 - If no y is found such that $P(x,y)$ is true the outer loop terminates as $\forall x \exists y P(x,y)$ has been shown to be false.

$\forall x \exists y P(x,y)$ is true if the outer loop ends after stepping through each x .

If the domains of the variables are infinite, then this process can not actually be carried out.

Order of Quantifiers

Examples:

1. Let $P(x,y)$ be the statement “ $x + y = y + x$.” Assume that U is the real numbers. Then $\forall x \forall y P(x,y)$ and $\forall y \forall x P(x,y)$ have the same truth value.
2. Let $Q(x,y)$ be the statement “ $x + y = 0$.” Assume that U is the real numbers. Then $\forall x \exists y Q(x,y)$ is true, but $\exists y \forall x Q(x,y)$ is false.

Questions on Order of Quantifiers₁

Example 1: Let U be the real numbers,

Define $P(x, y) : x \cdot y = 0$

What is the truth value of the following:

1. $\forall x \forall y P(x, y)$

2. $\forall x \exists y P(x, y)$

3. $\exists x \forall y P(x, y)$

4. $\exists x \exists y P(x, y)$

Questions on Order of Quantifiers₂

Example 2: Let U be the real numbers,

Define $P(x,y) : x / y = 1$

What is the truth value of the following:

1. $\forall x \forall y P(x, y)$

2. $\forall x \exists y P(x, y)$

3. $\exists x \forall y P(x, y)$

4. $\exists x \exists y P(x, y)$

Quantifications of Two Variables

Statement	When True?	When False
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y

Rules of Inference

Section 1.6

Section Summary₁

Valid Arguments

Inference Rules for Propositional Logic

Using Rules of Inference to Build Arguments

Rules of Inference for Quantified Statements

Building Arguments for Quantified Statements

Revisiting the Socrates Example

We have the two premises:

- “All men are mortal.”
- “Socrates is a man.”

And the conclusion:

- “Socrates is mortal.”

How do we get the conclusion from the premises?

The Argument

We can express the premises (above the line) and the conclusion (below the line) in predicate logic as an argument:

$$\frac{\forall x (Man(x) \rightarrow Mortal(x)) \quad Man(Socrates)}{\therefore Mortal(Socrates)}$$

We will see shortly that this is a valid argument.

Valid Arguments

We will show how to construct valid arguments in two stages; first for propositional logic and then for predicate logic. The rules of inference are the essential building block in the construction of valid arguments.

1. Propositional Logic

Inference Rules

2. Predicate Logic

Inference rules for propositional logic plus additional inference rules to handle variables and quantifiers.

Arguments in Propositional Logic

An **argument** in propositional logic is a sequence of propositions. All but the final proposition are called **premises**. The last statement is the **conclusion**.

The argument is valid if the premises imply the conclusion. An **argument form** is an argument that is valid no matter what propositions are substituted into its propositional variables.

If the premises are $p_1, p_2, \dots, p_n$ and the conclusion is q then $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is a tautology.

Inference rules are all argument simple argument forms that will be used to construct more complex argument forms.

Rules of Inference for Propositional Logic: Modus Ponens

$\frac{p \quad p \rightarrow q}{\therefore q}$	Corresponding Tautology: $(p \wedge (p \rightarrow q)) \rightarrow q$
--	---

Example:

Let p be “Texas is snowing.”

Let q be “Texans will lose power.”

“If Texas is snowing, then Texans will lose power.”

“Texas is snowing.”

“Therefore , Texans will lose power.”

Modus Tollens

$$\begin{array}{c} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

Corresponding Tautology:

$$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$$

Example:

Let p be “it is snowing.”

Let q be “I will study discrete math.”

“If it is snowing, then I will study discrete math.”

“I will not study discrete math.”

“Therefore , it is not snowing.”

Hypothetical Syllogism

$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$	Corresponding Tautology: $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
--	---

Example:

Let p be “it snows.”

Let q be “I will study discrete math.”

Let r be “I will get an A.”

“If it snows, then I will study discrete math.”

“If I study discrete math, I will get an A.”

“Therefore , If it snows, I will get an A.”

Disjunctive Syllogism

$$p \vee q$$

Corresponding Tautology:

$$\frac{\neg p}{\therefore q}$$

$$(\neg p \wedge (p \vee q)) \rightarrow q$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math or I will study English literature.”

“I will not study discrete math.”

“Therefore , I will study English literature.”

Addition

Corresponding Tautology:

$$\frac{p}{\therefore p \vee q}$$

$$p \rightarrow (p \vee q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will visit Las Vegas.”

“I will study discrete math.”

“Therefore, I will study discrete math or I will visit Las Vegas.”

Simplification

$$\frac{p \wedge q}{\therefore p}$$

Corresponding Tautology:

$$(p \wedge q) \rightarrow p$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math and English literature”

“Therefore, I will study discrete math.”

Conjunction

$$\frac{p}{\therefore p \wedge q}$$

Corresponding Tautology:

$$((p) \wedge (q)) \rightarrow (p \wedge q)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study English literature.”

“I will study discrete math.”

“I will study English literature.”

“Therefore, I will study discrete math and I will study English literature.”

Resolution

$$\frac{\begin{array}{l} p \vee q \\ \neg p \vee r \end{array}}{\therefore q \vee r}$$

Corresponding Tautology:

$$\left((\neg p \vee r) \wedge (p \vee q) \right) \rightarrow (q \vee r)$$

Example:

Let p be “I will study discrete math.”

Let q be “I will study databases.”

Let r be “I will study English literature.”

“I will study discrete math or I will study databases.”

“I will not study discrete math or I will study English literature.”

“Therefore, I will study databases or I will study English literature.”

Resolution

Decide satisfiability or validity by resolution rule.

Considering a set of clauses, use resolution rule to prove it is unsatisfiable:

$$1) \neg p(a) \vee \neg q(a) \vee r(a)$$

$$2) p(a)$$

$$3) \neg r(a)$$

$$4) \neg p(a) \vee q(a).$$

$$5) \neg q(a) \vee r(a), \text{ from 1) and 2)}$$

$$6) q(a), \text{ from 2) and 4)}$$

$$7) \neg q(a), \text{ from 3) and 5)}$$

from 6) and 7), we get an empty clause, or a contradictory.

Using the Rules of Inference to Build Valid Arguments

A *valid argument* is a sequence of statements. Each statement is either a premise or follows from previous statements by rules of inference. The last statement is called conclusion.

A valid argument takes the following form:

$$S_1$$
$$S_2$$
$$\cdot$$
$$\cdot$$
$$\cdot$$
$$S_n$$
$$\therefore C$$

Valid Arguments

Example 1: From the single proposition

$$p \wedge (p \rightarrow q)$$

Show that q is a conclusion.

Solution:

Step	Reason
1. $p \wedge (p \rightarrow q)$	Premise
2. p	Simplification using (1)
3. $p \rightarrow q$	Simplification using (1)
4. q	Modus Ponens using (2) and (3)

Valid Arguments

Example 2: Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

p : “It is sunny this afternoon.”

$\neg p \wedge q$

r : “We will go swimming.”

$r \rightarrow p$

t : “We will be home by sunset.”

$\neg r \rightarrow s$

q : “It is colder than yesterday.”

$s \rightarrow t$

s : “We will take a canoe trip.”

t

Valid Arguments

Example 2:

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Valid Arguments

Example 3: Show that the premises $(p \wedge q) \vee r$ and $r \rightarrow s$ imply $p \vee s$

Solution:

Distributive Law:

Simplification:

Logical Equivalence for $\rightarrow$:

Resolution:

Question

Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics.

You learned discrete mathematics.

Therefore, you did every problem in this book.

Fallacy of affirming the conclusion

Question

Is the following argument valid?

If you do every problem in this book, then you will learn discrete mathematics.

You did not do every problem in this book.

Therefore, you did not learn discrete mathematics.

Fallacy of denying the hypothesis

Handling Quantified Statements

Valid arguments for quantified statements are a sequence of statements. Each statement is either a premise or follows from previous statements by rules of inference which include:

1. Rules of Inference for **Propositional Logic**
2. Rules of Inference for **Quantified Statements**

Universal Instantiation (UI)

$$\frac{\forall x P(x)}{\therefore P(c)}$$

Example:

Our domain consists of all dogs and Fido is a dog.

“All dogs are cuddly.”

“Therefore, Fido is cuddly.”

Universal Generalization (UG)

$$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$$

Used often implicitly in Mathematical Proofs.

Existential Instantiation (EI)

$$\frac{\exists xP(x)}{\therefore P(c) \text{ for some element } c}$$

Example:

“There is someone who got an A in the course.”

“Let’s call her a and say that a got an A”

Existential Generalization (EG)

$$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$$

Example:

“Michelle got an A in the class.”

“Therefore, someone got an A in the class.”

Using Rules of Inference

Example 1: Using the rules of inference, construct a valid argument to show that
“John Smith has two legs”

is a consequence of the premises:

“Every man has two legs.” “John Smith is a man.”

Solution: Let $M(x)$ denote “ x is a man” and $L(x)$ “ x has two legs” and let John Smith be a member of the domain.

Valid Argument:

Step	Reason
1. $\forall x(M(x) \rightarrow L(x))$	Premise
2. $M(J) \rightarrow L(J)$	UI from (1)
3. $M(J)$	Premise
4. $L(J)$	Modus Ponens using (2) and (3)

Universal Modus Ponens

Universal Modus Ponens combines universal instantiation and modus ponens into one rule.

$$\frac{\begin{array}{l} \forall x(P(x) \rightarrow Q(x)) \\ P(a), \text{ where } a \text{ is a particular} \\ \text{element in the domain} \end{array}}{\therefore Q(a)}$$

This rule is used in the Socrates example.

Returning to the Socrates Example

$$\frac{\forall x (Man(x) \rightarrow Mortal(x)) \quad Man(Socrates)}{\therefore Mortal(Socrates)}$$

1. $\forall x (Man(x) \rightarrow Mortal(x))$, Premise
2. $Man(Socrates) \rightarrow Mortal(Socrates)$, UI from 1
3. $Man(Socrates)$, Premise
4. $Mortal(Socrates)$, Modus Ponens from 2 and 3

Using Rules of Inference

Example 2: Use the rules of inference to construct a valid argument showing that the conclusion

“Someone who passed the first exam has not read the book.”

follows from the premises

“A student in this class has not read the book.”

“Everyone in this class passed the first exam.”

Solution: Let $C(x)$ denote “ x is in this class,” $B(x)$ denote “ x has read the book,” and $P(x)$ denote “ x passed the first exam.”

First, we translate the premises and conclusion into symbolic form.

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Using Rules of Inference

Valid Argument:

Step	Reason
1. $\exists x(C(x) \wedge \neg B(x))$	Premise
2. $C(a) \wedge \neg B(a)$	EI from (1)
3. $C(a)$	Simplification from (2)
4. $\forall x(C(x) \rightarrow P(x))$	Premise
5. $C(a) \rightarrow P(a)$	UI from (4)
6. $P(a)$	MP from (3) and (5)
7. $\neg B(a)$	Simplification from (2)
8. $P(a) \wedge \neg B(a)$	Conj from (6) and (7)
9. $\exists x(P(x) \wedge \neg B(x))$	EG from (8)

$$\frac{\begin{array}{l} \exists x(C(x) \wedge \neg B(x)) \\ \forall x(C(x) \rightarrow P(x)) \end{array}}{\therefore \exists x(P(x) \wedge \neg B(x))}$$

Introduction to Proofs

Section 1.7

Section Summary₂

Mathematical Proofs

Forms of Theorems

Direct Proofs

Indirect Proofs

- Proof of the Contrapositive
- Proof by Contradiction

Proofs of Mathematical Statements

A **proof** is a valid argument that establishes the truth of a statement.

In math, CS, and other disciplines, informal proofs which are generally shorter, are generally used.

- More than one rule of inference are often used in a step.
- Steps may be skipped.
- The rules of inference used are not explicitly stated.
- Easier to understand and to explain to people.
- But it is also easier to introduce errors.

Proofs have many practical applications:

- verification that computer programs are correct
- establishing that operating systems are secure
- enabling programs to make inferences in artificial intelligence
- showing that system specifications are consistent

Definitions

A ***theorem*** is a statement that can be shown to be true using:

- definitions
- other theorems
- *axioms* (statements which are given as true)
- rules of inference

A ***lemma*** is a ‘helping theorem’ or a result which is needed to prove a theorem.

A ***corollary*** is a result which follows directly from a theorem.

Less important theorems are sometimes called *propositions*.

A ***conjecture*** is a statement that is being proposed to be true. Once a proof of a conjecture is found, it becomes a theorem. It may turn out to be false.

Forms of Theorems

Many theorems assert that a property holds for all elements in a domain, such as the integers, the real numbers, or some of the discrete structures that we will study in this class.

Often the universal quantifier (needed for a precise statement of a theorem) is omitted by standard mathematical convention.

For example, the statement:

“If $x > y$, where x and y are positive real numbers, then $x^2 > y^2$ ”

really means

“For all positive real numbers x and y , if $x > y$, then $x^2 > y^2$.”

Proving Theorems

Many theorems have the form: $\forall x(P(x) \rightarrow Q(x))$

To where c is an arbitrary element of the domain, $P(c) \rightarrow Q(c)$

By *universal generalization (UG)* the truth of the original formula follows.

So, we must prove something of the form: $p \rightarrow q$

Proving Conditional Statements: $p \rightarrow q$

Trivial Proof: If we know q is true, then

$p \rightarrow q$ is true as well.

“If it is raining then $1=1$.”

Vacuous Proof: If we know p is false then

$p \rightarrow q$ is true as well.

“If I am both rich and poor then $2 + 2 = 5$.”

[Even though these examples seem silly, both trivial and vacuous proofs are often used in mathematical induction, as we will see in Chapter 5)]

Background: Even and Odd Integers

Definition: The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k , such that $n = 2k + 1$. Note that every integer is either even or odd and no integer is both even and odd.

We will need this basic fact about the integers in some of the example proofs to follow. We will learn more about the integers in Chapter 4.

Proving Conditional Statements: $p \rightarrow q$

Direct Proof: Assume that p is true. Use rules of inference, axioms, and logical equivalences to show that q must also be true.

Example: Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Solution: Assume that n is odd. Then $n = 2k + 1$ for an integer k . Squaring both sides of the equation, we get:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2r + 1,$$

where $r = 2k^2 + 2k$, an integer.

We have proved that if n is an odd integer, then n^2 is an odd integer.

(marks the end of the proof. Sometimes **QED** is used instead.)

Proving Conditional Statements: $p \rightarrow q$

Definition: The real number r is *rational* if there exist integers p and q where $q \neq 0$ such that $r = p/q$

Example: Prove that the sum of two rational numbers is rational.

Solution: Assume r and s are two rational numbers. Then there must be integers p, q and also t, u such that

$$r = p / q, \quad s = t / u, \quad u \neq 0, \quad q \neq 0$$

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu} = \frac{v}{w}$$

$$\text{where } v = pu + qt$$

$$w = qu \neq 0$$

Thus, the sum is rational.

Proving Conditional Statements: $p \rightarrow q$

Proof by Contraposition: Assume $\neg q$ and show $\neg p$ is true also. This is sometimes called an *indirect proof* method. If we give a direct proof of $\neg q \rightarrow \neg p$ then we have a proof of $p \rightarrow q$.

Why does this work?

Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Solution: Assume n is even. So, $n = 2k$ for some integer k . Thus

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1) = 2j \text{ for } j = 3k + 1$$

Therefore $3n + 2$ is even. Since we have shown $\neg q \rightarrow \neg p$, $p \rightarrow q$ must hold as well. If n is an integer and $3n + 2$ is odd (not even), then n is odd (not even).

Proving Conditional Statements: $p \rightarrow q$

Exercise: Prove that for an integer n , if n^2 is odd, then n is odd.

Solution: Use proof by contraposition. Assume n is even (i.e., not odd). Therefore, there exists an integer k such that $n = 2k$. Hence,

$$n^2 = 4k^2 = 2(2k^2)$$

and n^2 is even (i.e., not odd).

We have shown that if n is an even integer, then n^2 is even. Therefore by contraposition, for an integer n , if n^2 is odd, then n is odd.

Proving Conditional Statements: $p \rightarrow q$

Proof by Contradiction: To prove q , assume $\neg q$ and derive a contradiction r , such that $\neg q \rightarrow r$ is true, which means $\neg q \rightarrow \mathbf{F}$, then it follows that the contrapositive $\mathbf{T} \rightarrow q$ also holds. How can we find such r ?

Example: Prove that if you pick 22 days from the calendar, at least 4 must fall on the same day of the week.

Solution: Assume that no more than 3 of the 22 days fall on the same day of the week. Because there are 7 days of the week, we could only have picked 21 days. This contradicts the assumption that we have picked 22 days.

Proving Conditional Statements: $p \rightarrow q$

Example: Use a proof by contradiction to give a proof that $\sqrt{2}$ is irrational.

Solution: Suppose $\sqrt{2}$ is rational. Then there exists integers a and b with $\sqrt{2} = a/b$, where $b \neq 0$ and a and b have no common factors. Then

$$2 = \frac{a^2}{b^2} \quad 2b^2 = a^2$$

Therefore a^2 must be even. If a^2 is even then a must be even. Since a is even, $a = 2c$ for some integer c . Thus, $2b^2 = 4c^2 \quad b^2 = 2c^2$

But then 2 must divide both a and b . This contradicts our assumption that a and b have no common factors. We have proved by contradiction that our initial assumption must be false and therefore $\sqrt{2}$ is irrational.

Proving Conditional Statements: $p \rightarrow q$

Exercise: Prove by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”.

Solution:

What is wrong with this?



“Proof” that $1 = 2$

Step

Reason

1. $a = b$

Premise

2. $a^2 = a \times b$

Multiply both sides of (1) by a

3. $a^2 - b^2 = a \times b - b^2$

Subtract b^2 from both sides of (2)

4. $(a - b)(a + b) = b(a - b)$

Algebra on (3)

5. $a + b = b$

Divide both sides by $a - b$

6. $2b = b$

Replace a by b in (5) because $a = b$

7. $2 = 1$

Divide both sides of (6) by b

Solution: Step 5. $a - b = 0$ by the premise and division by 0 is undefined.

What is wrong with this?

Show that n is an even integer whenever n^2 is an even integer.

Proof: Suppose that n^2 is even. Then $n^2 = 2k$ for some integer k . Let $n = 2l$ for some integer l . This shows that n is even.

Solution: $n = 2l$ for some integer l can't be assumed, this is what you're proving. (**begging the question**)

Proof Methods and Strategy

Section 1.8

Section Summary

Proof by Cases

Existence Proofs

- Constructive
- Nonconstructive

Disproof by Counterexample

Uniqueness Proofs

Proof Strategies

Proving Universally Quantified Assertions

Proof by Cases

To prove a conditional statement of the form:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

Use the tautology

$$\begin{aligned} & \left[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q \right] \leftrightarrow \\ & \left[(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q) \right] \end{aligned}$$

Each of the implications $p_i \rightarrow q$ is a *case*.

Proof by Cases

Example: Prove that if n is an integer, then $n^2 \geq n$.

Proof: We can split the proof into three cases:

1) $n = 0$

2) $n \geq 1$

3) $n \leq -1$

Proof by Cases

Example: Show that if x and y are integers and both $x \cdot y$ and $x + y$ are even, then both x and y are even.

Proof: Use a proof by contraposition. Suppose x and y are not both even. Then, one or both are odd. Without loss of generality, assume that x is odd. Then $x = 2m + 1$ for some integer m .

Case 1: y is even. Then $y = 2n$ for some integer n , so

$$x + y = (2m + 1) + 2n = 2(m + n) + 1 \text{ is odd.}$$

Case 2: y is odd. Then $y = 2n + 1$ for some integer n , so

$$x \cdot y = (2m + 1)(2n + 1) = 2(2m \cdot n + m + n) + 1 \text{ is odd.}$$

We only cover the case where x is odd because the case where y is odd is similar. The use phrase *without loss of generality* (WLOG) indicates this.

Existence Proofs

Proof of theorems of the form $\exists xP(x)$.

Constructive existence proof:

- Find an explicit value of c , for which $P(c)$ is true.
- Then $\exists xP(x)$ is true by Existential Generalization (EG).

Example: Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways:

Proof: 1729 is such a number since

$$1729 = 10^3 + 9^3 = 12^3 + 1^3$$

Nonconstructive Existence Proofs

In a **nonconstructive** existence proof, we assume no c exists which makes $P(c)$ true and derive a contradiction.

Example: Show that there exist irrational numbers x and y such that x^y is rational.

Proof: We know that $\sqrt{2}$ is irrational. Consider the number $\sqrt{2}^{\sqrt{2}}$. If it is *rational*, we have two irrational numbers x and y with x^y rational, namely $x = \sqrt{2}$ and $y = \sqrt{2}$. But if $\sqrt{2}^{\sqrt{2}}$ is *irrational*, then we can let $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$ so that $x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{(\sqrt{2}\sqrt{2})} = \sqrt{2}^2 = 2$.

Counterexamples

Recall $\exists x \neg P(x) \equiv \neg \forall x P(x)$.

To establish that $\neg \forall x P(x)$ is true (or $\forall x P(x)$ is false) find a c such that $\neg P(c)$ is true or $P(c)$ is false.

In this case c is called a *counterexample* to the assertion $\forall x P(x)$.

Example: “Every positive integer is the sum of the squares of 3 integers.” The integer 7 is a counterexample. So the claim is false.

Uniqueness Proofs

Some theorems assert the existence of a unique element with a particular property, $\exists!x P(x)$. The two parts of a *uniqueness proof* are

- *Existence*: We show that an element x with the property exists.
- *Uniqueness*: We show that if x and y both have the desired property, then $x=y$.

Example: Show that if a and b are real numbers and $a \neq 0$, then there is a unique real number r such that $ar + b = 0$.

Solution:

- *Existence*: The real number $r = -b/a$ is a solution of $ar + b = 0$ because $a(-b/a) + b = -b + b = 0$.
- *Uniqueness*: Suppose that s is a real number such that $as + b = 0$. Then $ar + b = as + b$, where $r = -b/a$. Subtracting b from both sides and dividing by a shows that $r = s$.

Proof Strategies for proving $p \rightarrow q$

Choose a method.

1. First try a direct method of proof.
2. If this does not work, try an indirect method (e.g., try to prove the contrapositive).

For whichever method you are trying, choose a strategy.

1. First try *forward reasoning*. Start with the axioms and known theorems and construct a sequence of steps that end in the conclusion. Start with p and prove q , or start with $\neg q$ and prove $\neg p$.
2. If this doesn't work, try *backward reasoning*. When trying to prove q , find a statement p that we can prove with the property $p \rightarrow q$.

Backward Reasoning

Example: Given two distinct positive real numbers x and y , prove their arithmetic mean is always greater than their geometric mean.

Proof:

Universally Quantified Assertions₁

To prove theorems of the form $\forall x P(x)$, assume x is an arbitrary member of the domain and show that $P(x)$ must be true. Using UG it follows that $\forall x P(x)$.

Example: An integer x is even if and only if x^2 is even.

Solution: The quantified assertion is

$$\forall x [x \text{ is even} \leftrightarrow x^2 \text{ is even}]$$

We assume x is arbitrary.

Recall that $P \leftrightarrow q$ is equivalent to $(P \rightarrow q) \wedge (q \rightarrow p)$

So, we have two cases to consider. These are considered in turn.

Continued on next slide →

Universally Quantified Assertions₂

Case 1. We show that if x is even then x^2 is even using a direct proof (the *only if* part or *necessity*).

If x is even then $x = 2k$ for some integer k .

Hence $x^2 = 4k^2 = 2(2k^2)$ which is even since it is an integer divisible by 2.

This completes the proof of case 1.

Case 2 on next slide →

Universally Quantified Assertions₃

Case 2. We show that if x^2 is even then x must be even (the *if* part or *sufficiency*). We use a proof by contraposition.

Assume x is not even and then show that x^2 is not even.

If x is not even then it must be odd. So, $x = 2k + 1$ for some k .

$$\text{Then } x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

which is odd and hence not even. This completes the proof of case 2.

Since x was arbitrary, the result follows by UG.

Therefore we have shown that x is even if and only if x^2 is even.